

37495 Statistical Design and Models for Evaluation Studies

Computer Lab 4

Question 1

The data below are the final weights of chicks (in grams) at six weeks, after having been fed on one of six diets. The trial was designed to study the comparative feeding values of six protein supplements. Ten chicks were used per lot. The birds were all of the same sex and age and from the same flock. All rations had the same level of crude protein (casein is a milk protein and forms the basis of cheese). All birds receiving each ration were housed together. Record was kept of the amount of feed eaten by lots only.

Table 1: Weights of chicks after being fed various diets

protein	1	2	3	4	5	6	7	8	9	10
horsebean	179	160	136	227	217	168	108	124	183	140
linseed	309	229	181	141	260	203	122	169	213	257
soybean	243	230	248	327	329	250	180	271	316	267
sunflower	423	340	392	339	341	226	320	295	334	322
meat	325	257	303	315	380	153	263	242	206	344
casein	368	390	379	260	404	318	352	359	216	222

- Horsebean - (Vegetable) (Bean)
- Linseed - (Vegetable) (Oil)
- Soybean - (Vegetable) (Oil)
- Sunflower - (Vegetable) (Oil)
- Meat - (Animal) (Meat)
- Casein - (Animal) (Dairy)

- a) With 0.05 significance level, determine if there is some treatment effect, using relevant plots to assess whether the conditions necessary for your conclusion to hold are met.
- b) Using Tukey's method to limit family-wise error rate to at most 0.05, identify the treatment pairs that have significantly different effects from each other.
- c) Using Dunnett's method to limit family-wise error rate to at most 0.05, identify the treatments that have significantly different effects from linseed oil meal.
- d) Using 0.05 significance level, determine if there is a significantly different effect between vegetable and animal proteins, making sure to write down the hypotheses that are being tested.
- e) Without suggesting extra factors should be included, identify two improvements that could be made to the design of this experiment, giving reasons for your answer.

Question 2

An experiment was run to compare the effects of auditory and visual cues on the speed of response of a human subject. A stimulus was presented to each subject on a computer and the reaction time required for the subject to press a key was recorded. The cue was either auditory or visual was presented 5, 10, or 15 seconds before the stimulus, with the corresponding reaction time recorded below.

Table 2: Reaction time for auditory and visual cues

treatment	1	2	3
Auditory 5	0.168	0.170	0.181
Auditory 10	0.167	0.182	0.187
Auditory 15	0.202	0.198	0.236
Visual 5	0.257	0.279	0.269
Visual 10	0.283	0.235	0.260
Visual 15	0.240	0.281	0.258

- a) Plot the data and identify any potentially statistically-significant differences between treatments (or groups of treatments).
- b) Using 0.05 significance level, perform a single hypothesis test to determine whether the treatments have a statistically-significant effect. Check whether the conditions necessary for your analysis hold.
- c) Define a set of contrasts (or the contrast weights) and corresponding null hypotheses that would allow you to determine statistically-different effects between:
 - 1) auditory and visual treatments
 - 2) 5 second lag treatments
 - 3) 10 second lag treatments
 - 4) 15 second lag treatments.
- d) Using Scheffe's method to limit family-wise error rate to at most 0.05, test the significance of the contrasts you defined in part c).